Unit 14 Haloalkanes and Haloarenes



Francois Auguste Victor Grignard

He a French chemist was awarded the noble prize for chemistry in 1912. He prepared and realised the usefulness of organo magnesium compounds. He took maths degree and eventually switched over organic chemistry while attempting to find the catalyst for the process of Methylation.

Learning Objectives

After learning this unit, students will be able to

- classify the various organic halo compounds.
- name the organic halo compounds according to IUPAC system.
- recognise the nature of C-X bond.
- describe the general methods of preparation of haloalkanes and haloarenes
- explain the physical and chemical properties of haloalkanes and haloarenes.
- describe the mechanism of nucleophilic substitution and elimination reactions of halo organics
- explain the preparation and synthetic uses of grignard reagent.
- highlight the uses of poly halogen compounds.
- comprehend the environmental effect of poly halogen compounds.

14.1 INTRODUCTION

In the previous unit we learnt about the chemistry of hydrocarbons. In this unit us learn about organic compounds containing halogens. When one or more hydrogen atoms of aliphatic or aromatic hydrocarbons are replaced by the corresponding number of halogens like fluorine, chlorine, bromine

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